5.5. **Weibull marginal models.** We equip the open half-line $\mathbb{X} = \mathbb{Y} =]0, \infty[$ with the Lebesgue measure $\lambda(dx) = \nu(dx) = 1_{]0,\infty[}(x)dx$ and consider the Weibull marginal measure

$$\lambda_U(dx) = \frac{a_u + 1}{b_u} \left(\frac{x}{b_u}\right)^{a_u} e^{-(x/b_u)^{a_u+1}} \lambda(dx)$$

with potential

$$U(x) = c_u + (x/b_u)^{a_u+1} + a_u \log \frac{1}{x}, \quad \text{where} \quad c_u := \log \frac{b_u}{a_u + 1} + a_u \log b_u$$

and $a_u, b_u > 0$. Note that (4.21) holds for any $0 < \delta < 1/2$ and the probability measures λ_U and λ_V have finite p_u and p_v absolute moments $m_u^{p_u}$ and $m_v^{p_v}$ for any p_u, p_v. We define similarly λ_V with some parameters $a_v, b_v > 0$.

Consider a continuous function $W(x, y)$ satisfying (5.2) for some parameters $c_w \in \mathbb{R}$ as well as $p_u, p_v > 0$. In this context, we have

$$W^V(x) \leqslant c_w + \varsigma_u \, x^{p_u} + \varsigma_v \, m_v^{p_v} \quad \text{with} \quad m_v^{p_v} := \int_0^\infty x^{p_u} \lambda_V(dx) < \infty.$$

This implies that

$$\delta U(x) - W^V(x) \geqslant \bar{c}_u + \delta a_u \log \frac{1}{x} + \frac{\delta}{b_u^{a_u+1}} \, x^{a_u+1} - \varsigma_u \, x^{p_u} \quad \text{with} \quad \bar{c}_u := \delta c_u - c_w - \varsigma_v \, m_v^{p_v}$$

Assume that (W_U, W^V) are both continuous functions. In this situation, $\mathcal{U}_\delta$ has compact sub-level sets with $0 < \delta < 1/2$ as soon as

$$a_u + 1 > p_u \quad \text{or} \quad \left(p_u = a_u + 1 \quad \text{and} \quad \varsigma_u \, b_u^{a_u+1} < \delta < 1/2\right).$$

The r.h.s. condition requires that $\varsigma_u < b_u^{-(a_u+1)}/2$. The analysis of these Weibull models now follows the same line of arguments as the ones we used for the Beta models.

5.6. **Exponential marginal models.** We equip the half-line $\mathbb{X} = \mathbb{Y} = [0, \infty[$ with the Lebesgue measure $\lambda(dx) = 1_{[0,\infty[}(x)dx$ and consider the exponential marginal measures

$$\lambda_U(dx) = \tau_u \, e^{-\tau_u x} \, \lambda(dx) \quad \text{and} \quad \lambda_V(dx) = \tau_v \, e^{-\tau_v x} \, \lambda(dx) \quad \text{with} \quad \tau_u \wedge \tau_v > 0$$

for which we have the potentials

$$U(x) = c_u + \tau_u x \quad \text{and} \quad V(x) = c_v + \tau_v x \quad \text{with} \quad (c_u, c_v) := (-\log \tau_u, -\log \tau_u).$$

Note that the functions $x \in [0, \infty[\mapsto \tau_\iota x \in [0, \infty[$ with $\iota \in \{u, v\}$ have compact sub-level sets. Let $\mathcal{Q}(x, dy)$ be the exponential transition defined for any $x \in [0, \infty[$ by

$$\mathcal{Q}(x, dy) = \frac{\varsigma}{1 + (1 - e^{-\lambda x})} \, e^{-\varsigma|x-y|} \, \lambda(dy) \quad \text{with } 0 < \varsigma < (\tau_u \wedge \tau_v)/2$$

which yields the cost function

$$W(x, y) \leqslant c_w + \varsigma|x - y| \leqslant c_w + \varsigma \, x + \varsigma \, y, \quad \text{with } c_w := \log(2/\varsigma) \, .$$

In this context, condition $\mathcal{H}'_\delta(U, V)$ holds for any $(\varsigma/\tau_u) \vee (\varsigma/\tau_v) < \delta < 1/2$.

5.7. **Semi-compact models.** In many practical situations of interest, the target state spaces are different. Next we illustrate this situation with a semi-discrete model and a semi-compact models.

Semi-discrete model. Consider a finite or countable subset $\mathbb{X} \subset \mathbb{R}^d$ equipped with the counting measure λ and the discrete topology. In this context, the potential function U is a function

$$U \;:\; x \in \mathbb{X} \mapsto U(x) \in [0, \infty[\quad \text{such that} \quad \sum_{x \in \mathbb{X}} e^{-U(x)} = 1.$$

Let $\nu_V(dy) = e^{-V(y)}\, dy$ be a Boltzmann-Gibbs probability measure on $\mathbb{R}^d$ and assume that (U, V, W) are continuous functions satisfying the polynomial growth conditions (5.2). In this context, the sub-level sets of U are finite, thus compact in $\mathbb{X}$.

Following the proof of Proposition 5.1 these conditions ensure that $\mathcal{H}'_\delta(U, V)$ holds for some $0 < \delta < 1/2$ as soon as condition (5.3) is satisfied. The analysis of heavy tailed transitions follows word-for-word the same lines of arguments as in Section 5.2, this it is skipped.

Semi-compact model. Consider the uniform measure λ_U on a compact space $\mathbb{X} \subset \mathbb{Y} = \mathbb{R}^d$ associated with a null potential $U = 0$, which yields $\lambda_U = \lambda$. Let ν_V be a centered Gaussian distribution with unit variance on $\mathbb{Y}$ and assume that the transition potential is chosen so that

$$W(x, y) \leqslant w(x) + a\, \|y\|^p \Longrightarrow W_U(y) \leqslant \lambda(w) + b\, \|y\|^p$$

for some continuous function $w(x)$ and some parameters $a \leqslant 2$ and $p \leqslant 1$. These models satisfy $\mathcal{H}'_\delta(U, V)$ as soon as

$$(p = 2 \text{ and } b < \delta/2) \quad \text{or} \quad (p < 2) \tag{5.5}$$

We check this claim using the fact that

$$(5.5) \Longrightarrow \delta V(y) - W_U(y) \geqslant c + \frac{\delta}{2}\, \|y\|^2 - b\, \|y\|^p \stackrel{\|y\| \to \infty}{\longrightarrow} +\infty \quad \text{for some } c \in \mathbb{R}.$$

5.8. **Linear-Gaussian models.** Consider the state spaces $\mathbb{X} = \mathbb{Y} = \mathbb{R}^d$, for some integer $d \geqslant 1$ and the Lebesgue measure $\lambda(dz) = \nu(dz) = dz$. Hereafter, points x in the Euclidean space $\mathbb{X} = \mathbb{R}^d$ are represented by d-dimensional column vectors (or, equivalently, by $d \times 1$ matrices).

Let $\mathcal{G}l_d$ denote the general linear group of $(d \times d)$-invertible matrices, and $\mathcal{S}_d^+ \subset \mathcal{G}l_d$ the subset of positive definite matrices. We sometimes use the Löwner partial ordering notation $v_1 \geqslant v_2$ to mean that a symmetric matrix $v_1 - v_2$ is positive semi-definite (equivalently, $v_2 - v_1$ is negative semi-definite), and $v_1 > v_2$ when $v_1 - v_2$ is positive definite (equivalently, $v_2 - v_1$ is negative definite). Given $v \in \mathcal{S}_d^+$ we denote by $v^{1/2}$ the principal (unique) symmetric square root.

Denote by $\nu_{m,\sigma}$ the Gaussian distribution on $\mathbb{R}^d$ with mean $m \in \mathbb{R}^d$ and covariance matrix $\sigma \in \mathcal{S}_d^+$. In addition, let g_σ denote the probability density function of the distribution $\nu_{0,\sigma}$, with covariance matrix $\sigma \in \mathcal{S}_d^+$. Consider the target probability measures $(\lambda_U, \nu_V) = (\nu_{m,\sigma}, \nu_{\overline{m},\overline{\sigma}})$ for some given parameters

$$(m, \overline{m}) \in (\mathbb{R}^d \times \mathbb{R}^d) \quad \text{and} \quad (\sigma, \overline{\sigma}) \in (\mathcal{S}_d^+ \times \mathcal{S}_d^+).$$

Equivalently, the log-density functions (U, V) are defined by

$$U(x) = -\log g_\sigma(x - m) \quad \text{and} \quad V(y) := -\log g_{\overline{\sigma}}(y - \overline{m}). \tag{5.6}$$

The linear Gaussian transition $\mathcal{Q}$ is defined by (1.2), with the potential

$$W(x, y) = -\log g_\tau(y - (\alpha + \beta x)) \tag{5.7}$$

for some given $(\alpha, \beta, \tau) \in \left(\mathbb{R}^d \times \mathcal{G}l_d \times \mathcal{S}_d^+\right)$. In this context, we have $\jmath(\mathcal{Q}) = 0$ and Hilbert projective techniques cannot be applied. However, a refined exponential stability analysis of the Sinkhorn algorithm for Gaussian models has been developed in [2]. For instance, following Remark 5.4 in [2], there exists some $n_0 \geqslant 1$ such that for any $n \geqslant n_0$ we have the estimate

$$\|\pi_{2n} - \nu_{\overline{m},\overline{\sigma}}\|_{\mathsf{tv}} \leqslant c\, \rho^n \quad \text{with} \quad \rho = (1 + \lambda_{\mathsf{min}}(\Delta_\varpi))^{-1} < 1. \tag{5.8}$$

The matrix Δ_ϖ in expression (5.8) above is defined as

$$\Delta_\varpi := \frac{\varpi}{2} + \left(\varpi + \left(\frac{\varpi}{2}\right)^2\right)^{1/2} \geqslant \varpi + (I + \varpi^{-1})^{-1} \tag{5.9}$$

with

$$\varpi := (\gamma\,\gamma')^{-1} \quad \text{and} \quad \gamma := \overline{\sigma}^{1/2}\,\tau^{-1}\beta\,\sigma^{1/2}.$$

The drift-minorization techniques developed in this paper also enable us to obtain exponential convergence on weighted Banach spaces.

Proposition 5.2. *Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S}^\circ)$ is satisfied with the Lyapunov functions*

$$\varphi(x) := 1 + (x-m)'\sigma^{-1}(x-m) \quad \text{and} \quad \psi(y) := 1 + (y-\overline{m})'\overline{\sigma}^{-1}(y-\overline{m}). \tag{5.10}$$

A proof is provided in the Appendix A (on page 34).

The choice of the Lyapunov functions is not unique. For instance, Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S}^\circ)$ is also met for any $\delta > 0$ with the exponential Lyapunov functions

$$\begin{aligned} \varphi(x) &:= \exp\left(\delta\,((x-m)'\sigma^{-1}(x-m))^{1/2}\right) \quad \text{and} \\ \psi(y) &:= \exp\left(\delta\,((y-\overline{m})'\,\overline{\sigma}^{-1}(y-\overline{m}))^{1/2}\right). \end{aligned} \tag{5.11}$$

There also exists some $\delta_0 > 0$ such that $\mathcal{D}_{\varphi,\psi}(\mathcal{S}^\circ)$ is met for any $0 < \delta < \delta_0$ with the exponential Lyapunov functions

$$\begin{aligned} \varphi(x) &:= \exp\left(\delta\,(x-m)'\sigma^{-1}(x-m)\right) \quad \text{and} \\ \psi(y) &:= \exp\left(\delta\,(y-\overline{m})'\overline{\sigma}^{-1}(y-\overline{m})\right). \end{aligned} \tag{5.12}$$

A detailed proof of the above assertions is provided in Appendix A (on page 34).

We end this section with an equivalent formulation of condition $\mathcal{H}_\delta(U,V)$ in terms of the parameters of the Gaussian potentials (5.6) and (5.7).

Proposition 5.3. *Condition $\mathcal{H}_\delta(U,V)$ is satisfied for the Lyapunov functions in (5.12) if, and only if,*

$$\beta'\tau^{-1}\beta < (\delta \wedge (1-\delta))\,\sigma^{-1} \quad \textit{and} \quad \tau^{-1} < (\delta \wedge (1-\delta))\,\overline{\sigma}^{-1}. \tag{5.13}$$

A proof is provided in Appendix E (on page 48).

5.9. **Finite mixture models.** Consider some (fully supported) discrete probability measures ξ_0 and ξ_1 on some some finite states E_0 and E_1 and some collection of probability measures $(\lambda_{U_i})_{i\in E_0}$ and $(\nu_{V_j})_{j\in E_1}$ for some continuous potential functions $U_i : x \in \mathbb{X} \mapsto U(x) \in \mathbb{R}$ and $V_j : y \in \mathbb{X} \mapsto V(y) \in \mathbb{R}$ indexed by $i \in E_0$ and $j \in E_1$. The mixture distributions λ_U and ν_V associated with these potential functions take the form

$$U = -\log \sum_{i\in E_0} \xi_0(i)\,e^{-U_i} \quad \text{and} \quad V = -\log \sum_{j\in E_1} \xi_1(j)\,e^{-V_j}. \tag{5.14}$$

Proposition 5.4. *Condition $\mathcal{H}_\delta(U,V)$ is satisfied for some $\delta \in]0,1[$ if conditions $\mathcal{H}_\delta(U_i,V_j)$ are met for the same $\delta \in]0,1[$ and almost every $(i,j) \in (E_0 \times E_1)$.*

A proof is provided in Appendix E (on page 49).

Let us now consider the mixture model (5.14) with $\mathbb{X} = \mathbb{Y} = \mathbb{R}^d$, the potential $W(x,y)$ defined in Eq. (5.7) and the log-densities

$$-U_i(x) = \log g_{\sigma_i}(x-m_i) \quad \text{and} \quad -V_j(y) := \log g_{\overline{\sigma}_j}(y-\overline{m}_j), \tag{5.15}$$